HACK 101 WORKSHOP

CarolinaCon 2026 | Ash Harmon

HACK 101

1_

Overview

Workshop Content
Meeting Setup
Whoami

2_

Hacking

The many meanings
Hacker Culture
How does AI fit in?

3_

CTFs

CTF Styles
Challenge Types
Hands-On Examples
Tips & Tools

4_

CLI

CLI vs GUI
Shell
Terminal
OverTheWire

Meeting Info

Jitsi

- Live participation
- Chat monitored
- Questions answered
- Can unmute for Q&A

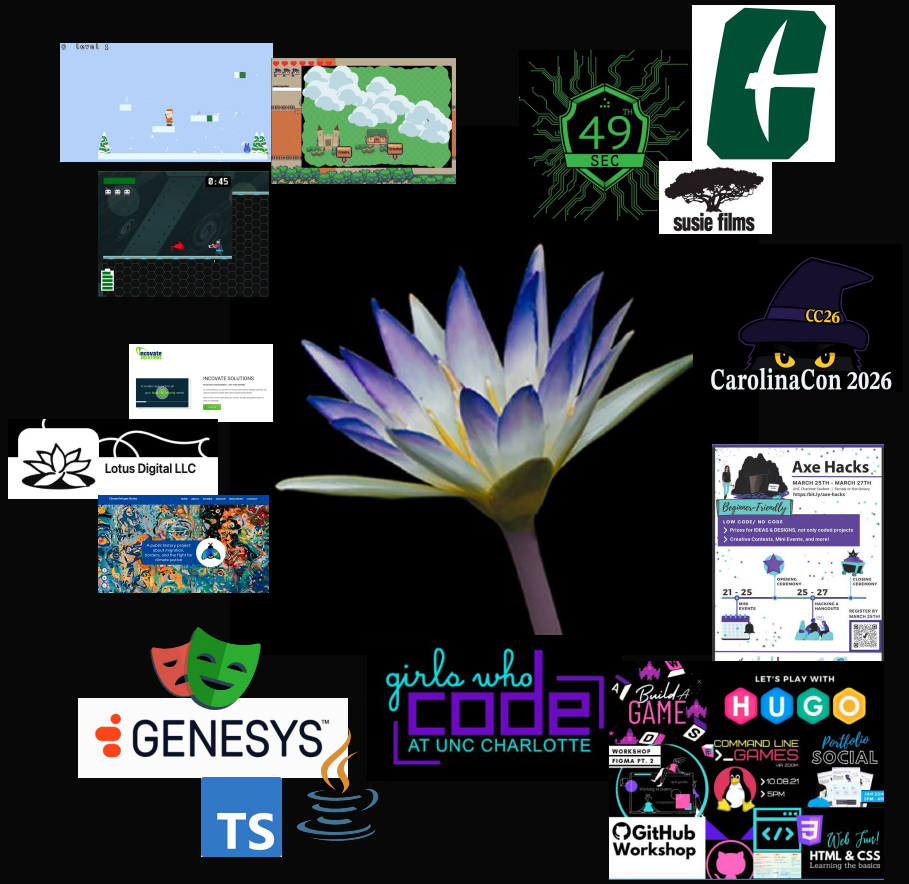
YouTube

- Live stream
- Chat not monitored
- Replay available afterwards

Register for link & passcode: **carolinacon.org/pages/join-us.html**

Jitsi Notes

- **Jitsi chat** for workshop-related things | **Discord** for general chat
- Workshop-related issues → **Jitsi chat**
- General tech / Jitsi join support → **#tech-support** channel in Discord
- Tips on using Jitsi → **join-us page** above



Ash

Multi-faceted mischief maker, creative dabbler, fumbling my way to growth

UNC Charlotte Grad

B.S. Computer Science - Cyber Security

M.S. Information Technology - Human-Computer Interaction

Career

(former) Genesys - Software Engineer in Test (SET)

(current) Lotus Digital LLC - Web Design & Devopment

Community

49th Security Division - member

Girls Who Code at UNC Charlotte - Media Chair

CarolinaCon - Co-Organizer

CONTACT INFO

@bluelotus03 - YouTube, Discord, GitHub

bluelotus03.tech → Updates in progress

IMPORTANT NOTES

- **Links & Resources**
  - These are starting points, aid in knowing topics to explore, learn about techniques / opinions / concepts
  - Not guarantees of 100% accurate or up-to-date info
- **Ethics & Legality**
  - Use skills / techniques / tools responsibly
  - There are legal lines that may sometimes be hard to know but are important to learn – do your own research before wielding your hackery in the real-world (don’t hack things you don’t own)
  - Enjoy legal / contained environments (like CTFs) in the meantime

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“He just couldn’t **hack** it”

“The ultimate life **hack**”

“They found a cool **hack**”

“They were **hacking** all weekend”

“**Hackers** created an ecofriendly solution to …”

“He **hacked** at it with …”

“**Hackers** brought down the server”

So what even is it?



"He just couldn't **hack** it"
- Couldn't make it work / figure it out / stick it out
- A server, a job, a problem

"They found a cool **hack**"
- Work around / new (often unintended) function
- An exploit, re-purposed item

"The ultimate life **hack**"
- Improve life quality / efficiency / enjoyment
- A habit, lifestyle, product

"They were **hacking** away all weekend"
- Hackathons, hacking conferences, chopping
- Creating projects / exploits, chopping up wood

"**Hackers** created an ecofriendly solution to …"
- Hackathon participants
- Apps, websites, hardware

"He **hacked** (at) it with …"
- Attempt to gain access / exploit, chopped
- A server / system / hardware, chopped wood

"**Hackers** brought down the server"
- CTF participants or real-world hackers
- People who gained access / ran exploits in real-world or competition systems

🙃 **Context is everything**

HACKING

A "**Hack**" in General

Creativity • Solve a Problem / Reach a Goal • Not always technical

Find a New Use / Purpose / Functioning of something (object/mentality/functionality)

"**Hacking**"

The act of working on a hack

... or destroying with an axe

Hackathons

- Competitive event with prizes for projects that meet specific requirements
- Involves technology (physical, digital)
- Build something (website, app)
- Solve a problem (recycling app)
- Reach a Goal (fastest/easiest to use app for ___)

Hacking Conferences

- Conference with talks, workshops, and other events (CTFs, Lockpicking, etc.)
- Capture the Flag (CTF) Competitions
  - Involves technology (physical, digital)
  - Build something (device, script)
  - Solve a problem (find info, gain access, etc.)
  - Reach a Goal (crack password, run exploit)

› The (OG) hacker culture is a subculture of individuals who enjoy—often in collective effort—the intellectual challenge of **creatively overcoming** the limitations of **software systems** or **electronic hardware** (mostly digital electronics), **to achieve** novel and clever **outcomes.**

HACKER CULTURE

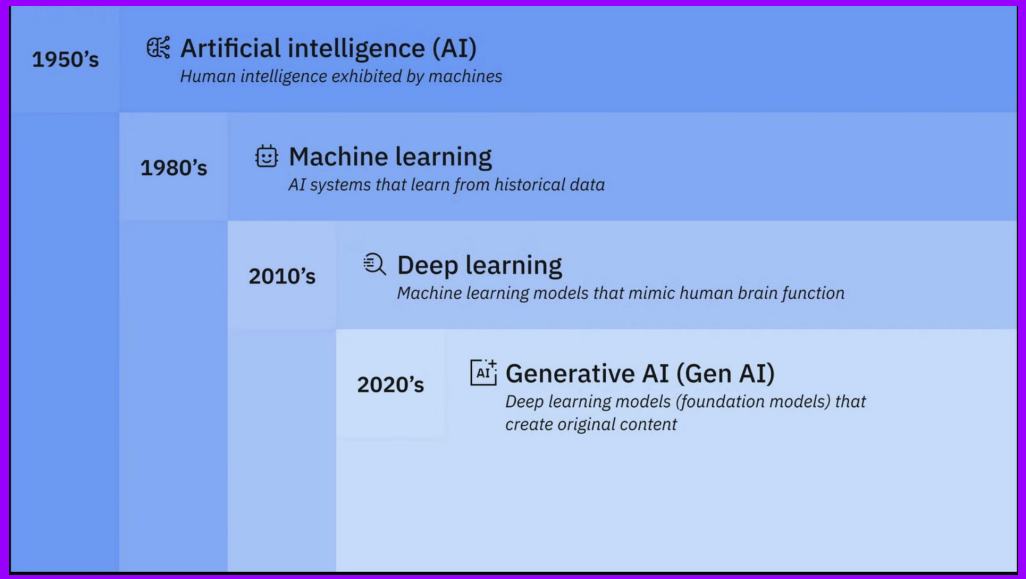
WHY BOTHER?

I got AI bro...



Artificial Intelligence (AI)

That's a whole other topic...



Learn More:

https://www.ibm.com/think/topics/artificial-intelligence

https://youtu.be/Uoz_osFtw68

https://youtu.be/mUw27wG7uFA

https://youtu.be/qYNweeDHiyU

https://www.ibm.com/think/topics/artificial-intelligence

https://aws.amazon.com/what-is/foundation-models/

https://x.com/curi0usJack/status/2024184571974000984

Generative AI (Gen AI)

Benefits

- Explore well-researched topics & potentially form new conclusions
- Agentic functionality to act on your behalf – more versatile than a script / program defined bot
- Find information and summarize it faster than you can yourself searching online resources

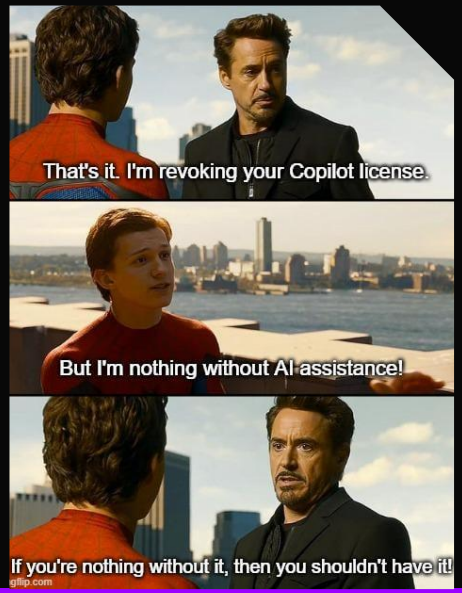
Limitations

- No guarantees about the resources it uses
- Not able to provide accurate new info it hasn't been trained on - could "hallucinate"
- Unable to determine what's correct without guiding prompts to help it arrive at that info – no "common sense" / judgement
- Relying on it for everything takes away the benefit of **you**

Example of some things it can currently do:

https://www.youtube.com/@JupiterBroadcasting/search?query=clanker

Why bother doing things myself?







JUDGEMENT

Think of AI tools like a Jr dev ...not a senior

It can assist you, offer perspective & research it gathered - save time ... but it should not run the show

VALUE

Every new technology cl;aims something, someone, some skill will be obselete

Your skills/curiosity/effort are still, and will always be, valuable

MENTALITY

Not everything has to be profitable or someone's definition of "productive"

Learning, and leveling up not only your skills but the way you think, is priceless

› The (OG) hacker culture is a subculture of individuals who **enjoy**—often in **collective effort**—the **intellectual challenge** of **creatively overcoming** the limitations of **software systems** or **electronic hardware** (mostly digital electronics), **to achieve** novel and clever **outcomes.**

The hacker mentality – being able to question things and take creative approaches is a huge part of why security professionals get paid

They question if something is actually secure, learn tips/tools/techniques for exploiting things from other's experiences (classes, workshops, blogs, etc.)

HACKER CULTURE

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Capture the Flag (CTF)

One of the live/hands-on ways to be a part of hacker culture

Provides a legal environment for people to hack real world systems that are intentionally vulnerable to help educate people in a gamified and/or group way → use skills responsibly

Jeopardy

- Most common (and our CTF format for CarolinaCon)
- Challenges typically split up into categories
- Earn points with correctly answered challenges

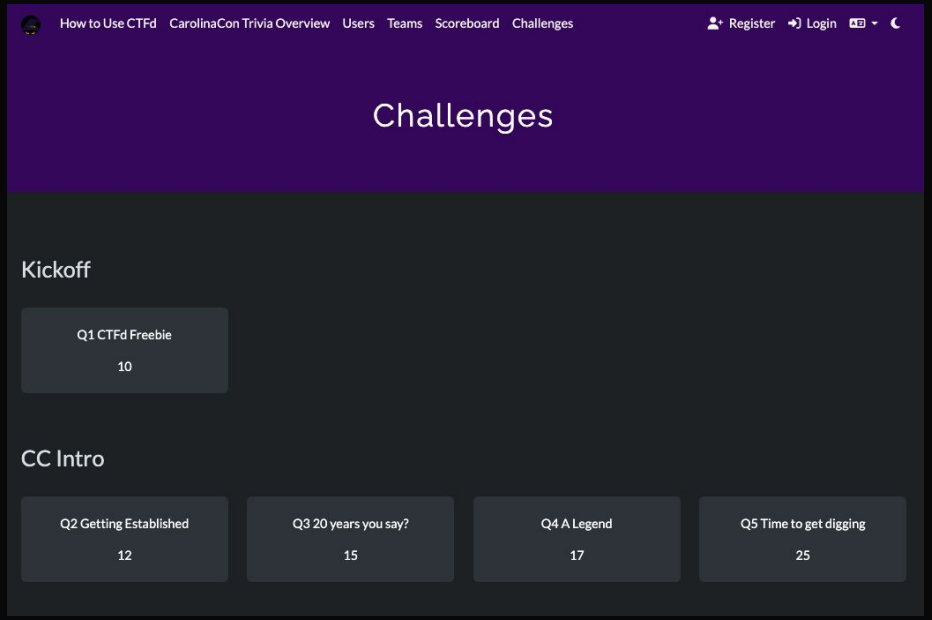
Attack/Defend

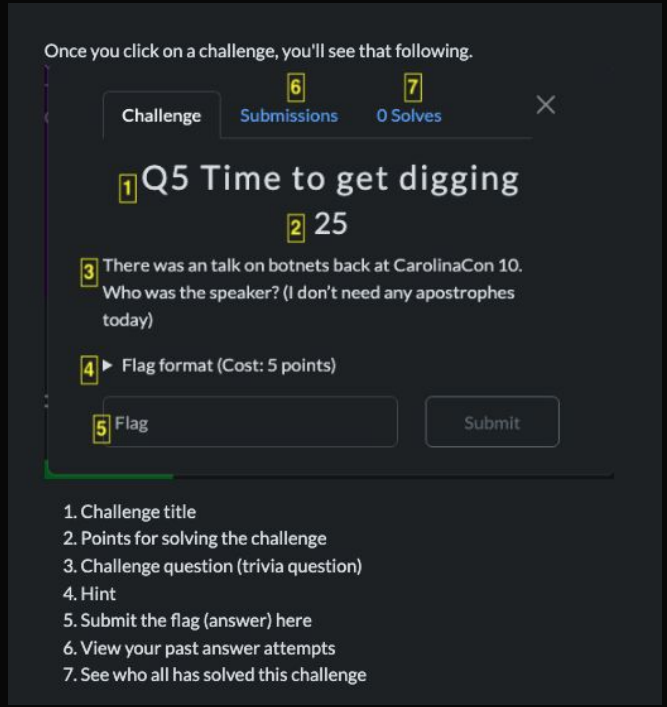
- Servers to defend
- Team tries to hack into other team (Offense)
- Other team defends (Defense)
- i.e. King of the Hill

JEOPARDY STYLE CTF

CarolinaCon - CTFd

`flag{somethingHere}`





TYPICAL CTF CATEGORIES:

OSINT / MISC

WEB EXPLOITATION (WEB)

CRYPTOGRAPHY

FORENSICS / STEGANOGRAPHY

REVERSE ENGINEERING (RE)

BINARY EXPLOITATION (PWN)

DON'T BE DISCOURAGED

Some challenges can be **intentionally deceitful** (i.e. using the wrong file extension)
→ there to encourage your curiosity and challenge your assumptions

More people are "lost" than you know – it's **part of the journey**

Be **realistic** with your expectations – the first time doing anything usually entails **99% research**
→ Don't jump in as a newbie to CLIs and be mad at yourself for not grasping a RE challenge

Ask for help / **Team up** with people

Approach things with **curiosity**, even if you end something still not understanding
→ Go to workshops / try CTFs / Read writeups / Watch tutorials - rinse and repeat, keep it **fun**

Difficulty: Beginner

OSINT / MISC

OSINT → **O**pen **S**ource **Int**elligence, intelligence produced by collecting and analyzing public information with the purpose of answering a specific intelligence question. – SANS

Involves Google dorking, metadata analysis, etc. – "finding info" however it is available

Information ≠ Intelligence
info = raw data intelligence = critical examination of info, knowing why it matters

Core Skills → Research Methods, Critical Thinking

Key Tools → Search Engines, exiftool, Python

Career Paths → Threat Intelligence Analyst

Learn More:

https://www.sans.org/blog/what-is-open-source-intelligence

https://www.recordedfuture.com/threat-intelligence-101/threat-analysis-techniques/google-dorks

https://medium.com/@irina0110/the-boundary-between-osint-and-reconnaissance-a-gray-area-27e91836c6b3

OSINT / MISC CHALLENGE

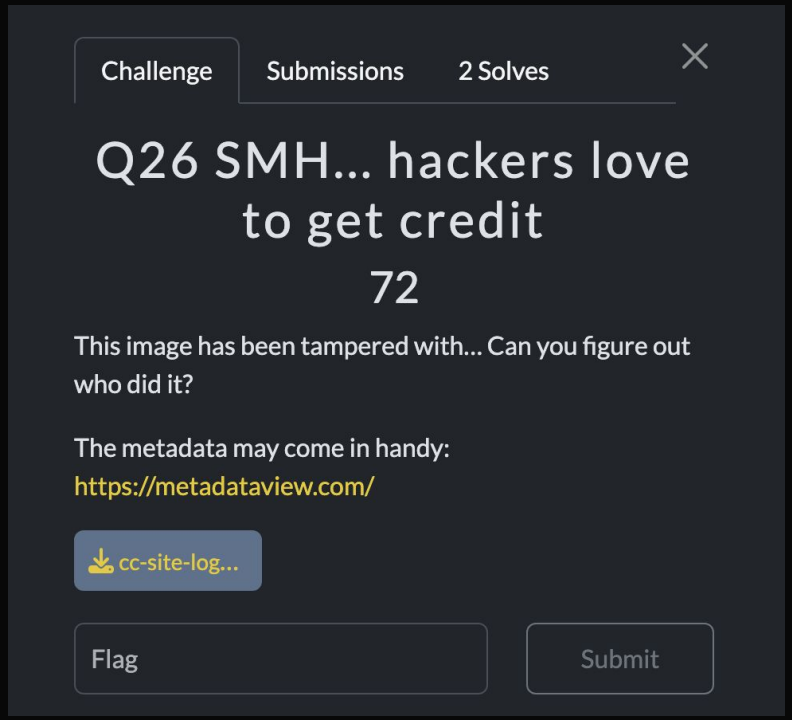
BASIC INFO COLLECTION



https://trivia.carolinacon.org/

OSINT / MISC CHALLENGE

METADATA ANALYSIS



Difficulty: Beginner

WEB EXPLOITATION (WEB)

Involves locating & exploiting vulnerabilities in a website or web app for various purposes (i.e. find flags for a CTF, gain unauthorized access to employee/customer info in real-world)

Flags could be found be inspecting the code HTML comment, using SQLi, directory brute forcing, etc.

Core Skills → HTML, HTTP, JavaScript

Key Tools → Burp Suite, Browser DevTools

Career Paths → Penetration Tester, AppSec Engineer

Learn More:

https://metactf.com/blog/ctf-101-series-what-is-web-exploitation/

WEB CHALLENGE

INSPECT



Challenge | Submissions | 0 Solves

Q27 Guess he has an official title

75

I bet the CarolinaCon creature will let you pet him if you address him correctly ...but man that's a long one

Flag

Submit

WEB CHALLENGE
INSPECT



Challenge | Submissions | 0 Solves

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75

I bet the CarolinaCon creature will let you pet him if you address him correctly ...but man that's a long one

Flag

Submit



Think he liked it… lol

WEB CHALLENGE

INSPECT w/ CONDITIONS



CRYPTOGRAPHY

Difficulty: Beginner to Intermediate

Involves identifying encoding (i.e. base64), cracking hashes, and known vulnerable cryptographic attacks

Core Skills → Math, Python, Pattern Recognition

Key Tools → CyberChef, Python (pycryptodome)

Career Paths → Cryptanalyst, Security Engineer

Learn More:

https://www.ibm.com/think/topics/cryptography

https://medium.com/@rudra-mehta/what-is-a-cipher-in-cryptography-be28238ceb58

CRYPTOGRAPHY CHALLENGE

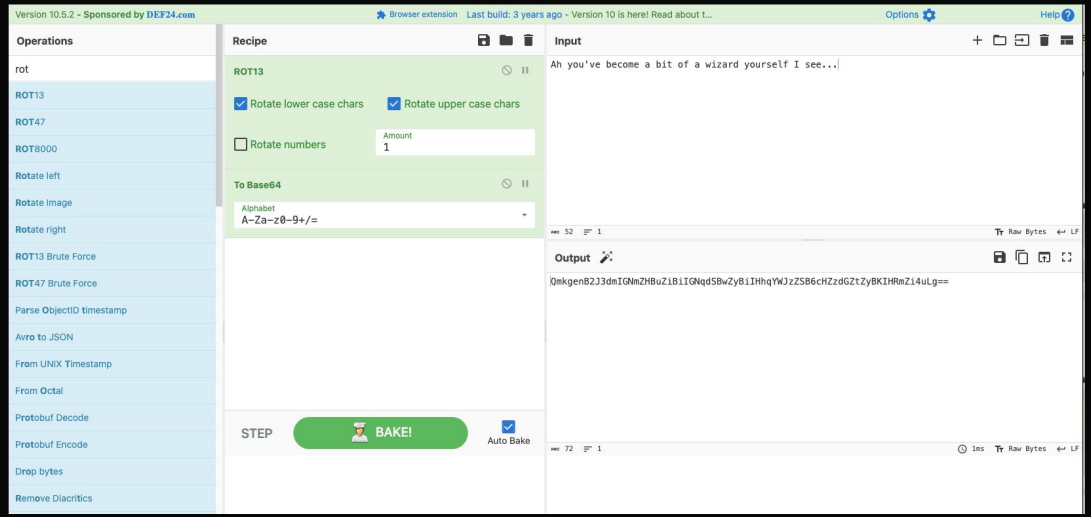
DECODE



CRYPTOGRAPHY CHALLENGE

ENCODE

https://cyberchef.org/



→ Try encoding your own sometime! :D

FORENSICS / STEGANOGRAPHY

Difficulty: Beginner to Intermediate

Computer forensics – also known as “digital forensics” or “computer forensics”, combines computer science and legal forensics to gather digital evidence

→ In a CTF, can involve recovering hidden, deleted, or encoded data from files, disk images, memory dumps, or network packets

→ OSINT is sometimes lumped into Forensics as well

Steganography – the practice of hiding an image, message, or file within something that isn't a secret

Core Skills → Linux CLI, Networking, File Systems

Key Tools → Wireshark, Volatility, binwalk

Career Paths → DFIR Analyst, Incident Responder

Learn More:

https://www.ibm.com/think/topics/computer-forensics

https://www.okta.com/identity-101/steganography/

FORENSICS EXAMPLE

WIRESHARK

Slides from an Intro to WireShark
→ tool for capturing live network traffic (packet sniffer)

https://github.com/49thSecurityDivision/Presentations/tree/master/2018-2019/EducationNights/Wireshark

https://www.wireshark.org/

REVERSE ENGINEERING (RE)

Difficulty: Intermediate to Advanced

Reverse engineering - in cybersecurity - the process of deconstructing compiled software, binaries, malware, or hardware systems to understand their internal logic, architecture, and behavior without access to the original source code

In CTFs, often involves deconstructing – or finding ways to deconstruct – an artifact / binary to understand how it executes

Core Skills → C Programming, Assembly Basics

Key Tools → Ghidra, GDB + pwndbg

Career Paths → Malware Analyst, Vulnerability Researcher

Learn More:
https://www.huntress.com/cybersecurity-101/topic/what-does-a-reverse-engineer-do-cybersecurity

RE EXAMPLE

BINARY MALWARE

CTF write-up from a binary malware challenge:

https://medium.com/@ad3mx0/we-have-a-binary-malware-dontrunmeso-51eeed0a76b2

BINARY EXPLOITATION (PWN)

Difficulty: Advanced

Involves (typically) connecting to raw sockets / ports and reverse engineering (when provided a program) to write custom exploits

Can also involve scripting something against a remote port / program to exploit it and solve for a flag

Core Skills → C, Memory Layout, Assembly

Key Tools → pwntools, GDB + pwndbg

Career Paths → Exploit Developer, Security Researcher

Learn More:
https://izzy.sh/posts/pwners-roadmap/

PWN EXAMPLE

BINARY EXPLOITATION

CTF write-up from a binary exploitation challenge:

https://github.com/vabhishek-me/picoctf-2017-write-up/blob/339f0b0487f47648a9e89357f679954eec20fc9c/Level 1/BINARY_EXPLOITATION/bash_loop.md

- Typically people use Linux for challenges (i.e. Kali Linux) – but can use any O.S.
  - Factors for which to O.S. to use – ease of workflow / setup, cost, availability & maintenance
  - Servers / machines you hack into could be Linux – good to know how to navigate through them
- Pick up a programming language & a scripting language
  - You can effectively “program” in bash (or other scripting languages) – but learning programming languages help you read exploits & better understand potential vulnerabilities you could come across in a live environment
- Get familiar with the CLI – you won’t always have a GUI when you test things
  - i.e. C2 frameworks, off the shelf exploits (metasploit or exploitdb), etc.

GENERAL
TIPS & TOOLS



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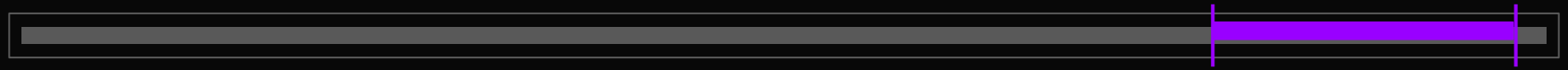
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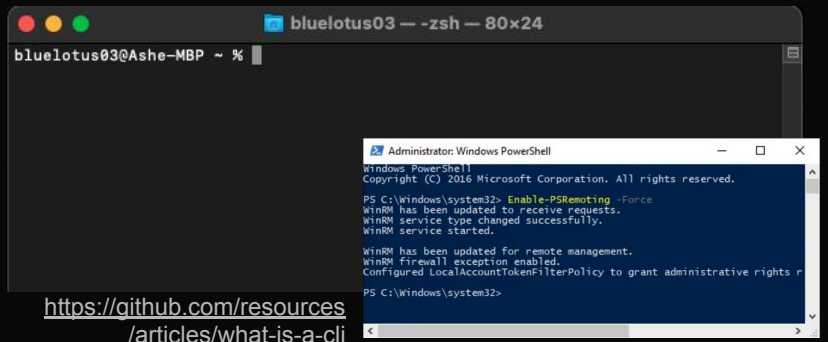
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Terminal
OverTheWire



CLI vs GUI

CLI

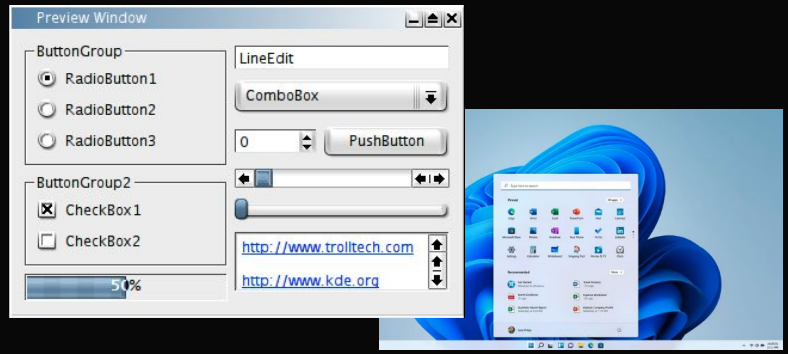
- Command Line Interface
- Uses **commands**
- Shell - interprets commands
- Terminal - application that launches your shell and CLI experience



https://github.com/resources/articles/what-is-a-cli

GUI

- Graphical User Interface
- Uses **mouse / clicks**
- GUI executes command / applications when you click on things



https://monovm.com/blog/what-is-gui/

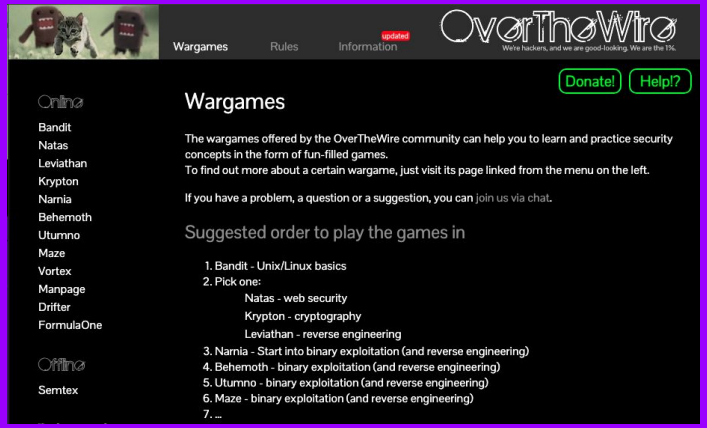
OVERTHEWIRE.ORG

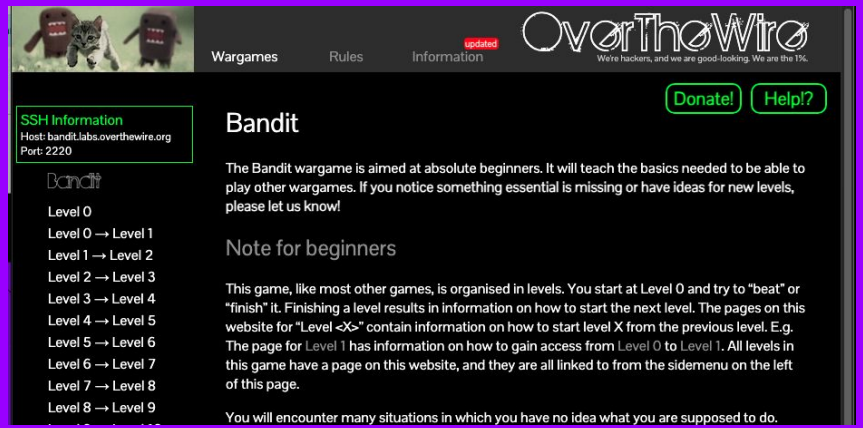
Windows: **Powershell**

Mac: **Terminal**

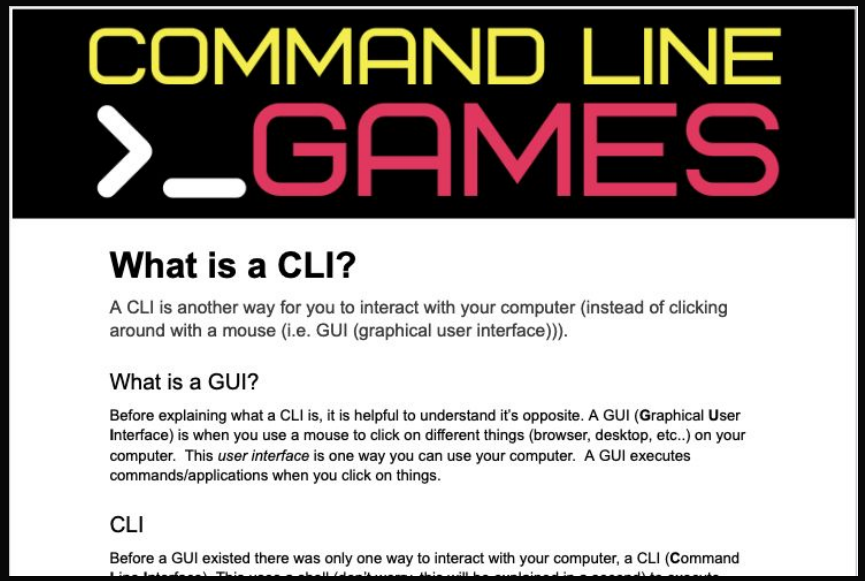
The OverTheWire community offers several different wargames that give you a fun way to learn and practice different security concepts.

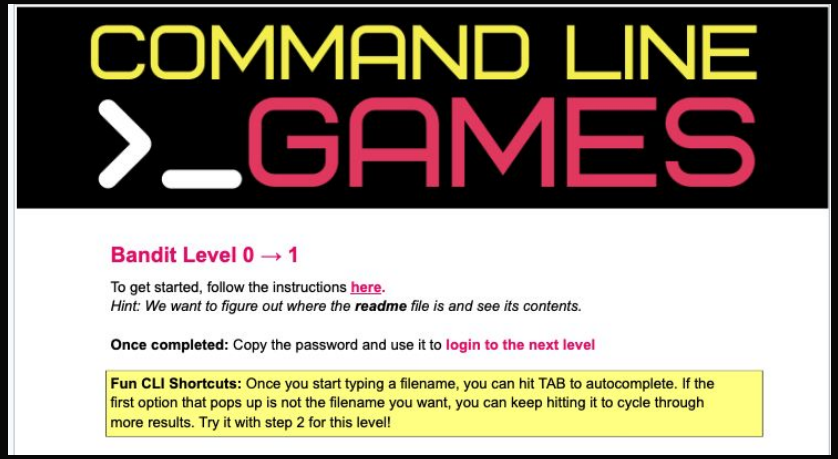
The **Bandit** wargame → to learn some of the basic commands on a Linux CLI
(recommend at least levels 0 to 5 to dip your toes in)





OverTheWire Tips





Command Line Games doc

Need Help? **#otw-wargames** channel in Discord

RESOURCES - hands on practice

- https://overthewire.org/wargames/bandit/ - Linux CLI wargames
- https://underthewire.tech/ - powershell wargames
- https://hackthebox.com/ (HTB) - larger community backing and/or well known
  - https://help.hackthebox.com/en/articles/5185687-gs-introduction-to-lab-access
  - https://ippsec.rocks/
- https://tryhackme.com/ - like HTB
- https://pwn.college/ - pwn challenges
- https://portswigger.net/web-security - learning web exploitation
- https://ctftime.org/ - upcoming CTFs
- https://www.antisyphontraining.com/ - affordable industry training

RESOURCES - online tools

Tools & resources around tools:

- https://cyberchef.org/
- https://www.jwt.io/ or https://jwt.ms/
- https://hacktricks.wiki/

Intentionally vulnerable applications:

- https://vwad.owasp.org/
- https://github.com/vavkamil/awesome-vulnerable-apps

RESOURCES - videos

- The Real Hacking Game Beginners Should Start With (CTF Explained)
- CTF Walkthrough with John Hammond
- BEGINNER Capture The Flag - PicoCTF 2021 001 "Obedient Cat" - good intro video & referenced briefly in previous video
  - picoCTF has migrated to CyLab Security Academy
- Beginner Friendly CTF Walkthrough - more advanced
  - Expects you to be familiar with CLI + networking
- Launch your cybersecurity career: IppSec's advice on how to become a skilled professional - specifically number 6 (but all are good)
  - Ippsec is a well known person in the cybersecurity community
- CTF Series - more advanced solving things & not as much explaining

RESOURCES - industry blogs (not CTFs)

- https://trustedsec.com/blog
- https://www.blackhillsinfosec.com/blog/
- https://redsiege.com/red-siege-blog/
- https://www.outflank.nl/blog/
- https://www.synacktiv.com/en/publications
- https://specterops.io/blog/
- https://www.secureideas.com/blog
- https://www.redteam.community/tradecraft
- https://www.netspi.com/blog/technical-blog/
- https://www.antisyphontraining.com/blogs/

THANK YOU!

carolinacon.org/pages/join-us.html